SmartCliff Learning Solutions

COE_KIOT_SDET-III

Core Java

Technical Assessment 13

Topic: Java Fundamentals [Basics, Control Flow Statements, Array, Functions]

Date: 18/02/2026

Time: 3:00 PM – 4:30 PM

Duration: 1.5 Hours

Total Marks: 50 Marks

PART A: Multiple Choice Questions (10 X 1 = 10)

PART B: Subjective Questions (40 Marks)

1. In Lucky draw event, a group of peoples participate, each participant is given a 4-digit random token starting from 1000 to 9999. Lucky draw is taken, a first prize of amount of 10000 is given for the token number whose sum of digits equal to 7, second prize amount of 5000 is given for the token number whose sum of digits is equal to 5, third prize amount of 3000 is given for the token number whose sum of digits is equal to 3.
   Sample Input and Output: **(8 Marks)**
   No of Participants: 10
   Token Number for 10 participants: [1234, 2345, 3456, 4567, 7891, 1000, 1200, 9999, 7777, 1224]
   First Prize is to token: 7891
   Second Prize is to token: 2345
   Third Prize is to token: 1200

2. "Secure Assets Private Ltd", a small company that deals with lockers has recently started manufacturing digital locks which can be locked and unlocked using PINs (passwords). You have been asked to work on the module that is expected to generate PINs using three input numbers. Assumptions: The three given input numbers will always consist of three digits each i.e. each of them will be in the range >=100 and <=999 **(8 Marks)**
   100 <= input1 <= 999
   100 <= input2 <= 999
   100 <= input3 <= 999
   Below are the rules for generating the PIN -
   - The PIN should be made up of 4 digits
   - The unit (ones) position of the PIN should be the least of the units position of the three input numbers
   - The tens position of the PIN should be the least of the tens position of the three input numbers
   - The hundreds position of the PIN should be the least of the hundreds position of the three input numbers

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- The thousands position of the PIN should be the maximum of all the digits in the three input numbers

Sample Input and Output

input1 = 123
input2 = 582
input3 = 175
then, PIN = 8122

3. A mathematician wants to check whether a given positive number is a "Perfect Number" or not. They input a positive integer, and the program determines if the number equals the sum of its proper divisors (factors excluding the number itself). The mathematician receives a message indicating whether the number is perfect or not. **(6 Marks)**

   Sample Input: Enter a positive integer: 6
   Sample Output: The given number is perfect
   Sample Input: Enter a positive integer: 65
   Sample Output: The given number is not perfect

4. Given a start number and end number, you need to display the numbers inclusive but "ding" for every 5th number and "dong" for every 10th number. **(6 Marks)**

   Inputs: Starting number and Ending number

   Outputs: Display "ding" for every 5th number and "dong" for every 10th number

   Sample Input: 11, 40

   Sample Output: 11 12 13 14 ding 16 17 18 19 dong 21 22 23 24 ding 26 27 28 29 dong 31 32 33 34 ding 36 37 38 39 dong

5. A teacher is collecting test scores from students for a final exam. The teacher enters the scores of each student one by one. The input stops when the teacher enters a negative number. The teacher wants to process the scores using a function based on certain conditions: **(10 Marks)**

   - Replace the score with 0 if it is even.
   - Replace the score with 1 if it is odd.
   - Replace the score with 2 if it is divisible by 8.
   - Replace the score with 3 if it ends with the digit 3.
   - Replace the score with 4 if it is divisible by 9.
   - If multiple rules apply to a number, the teacher wants to use the rule that replaces it with the highest number.

   The teacher then wants to see the original scores and the modified scores for further analysis.

   Sample Input: Enter the scores (enter a negative number to stop input):
   45 28 65 91 33 40 72 63 -1
   Sample Output: Scores before processing:
   45 28 65 91 33 40 72 63

   Scores after processing:
   4 0 1 1 3 0 2 4

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